

Range Minimum Queries

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1 Introduction

This project is my submission for the programming Project for the Advanced Data Structures course, Summer 2026. This project was implemented in C++, where the source file can be found under the directory `rmq-cpp`.

2 Methods

2.1 Naive

2.2 Precompute everything

2.3 Sparse Arrays

2.4 Segment Trees

2.5 Blocks

2.6 Cartesian Trees

3 Results

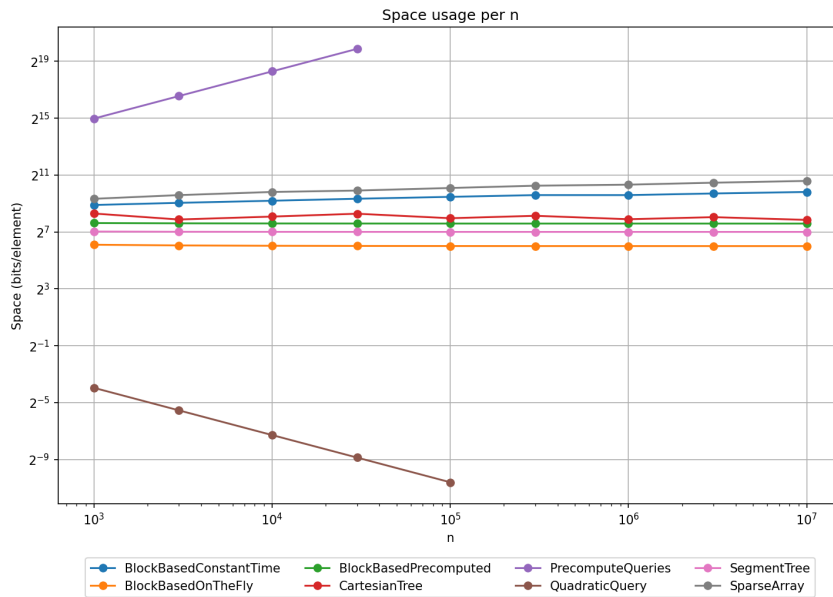


Figure 1 space usage

TODO

2 Range Minimum Queries

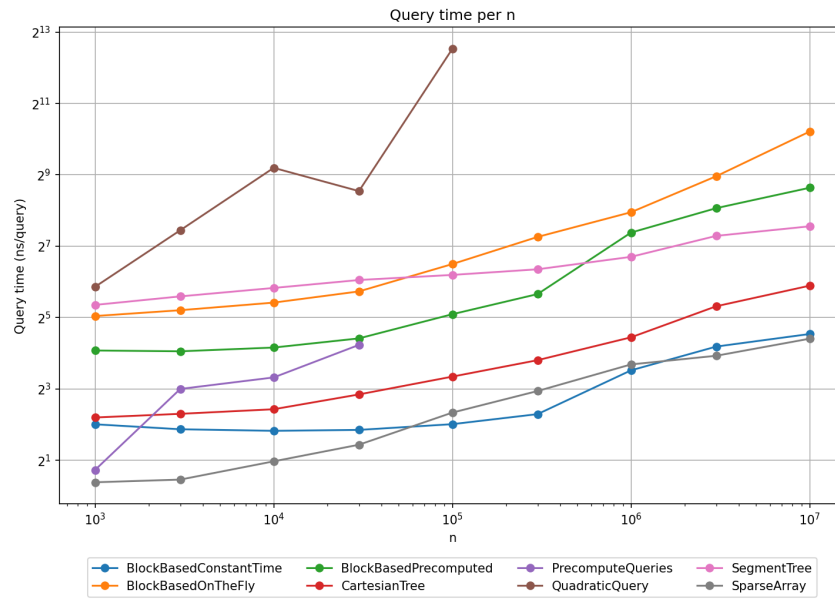


Figure 2 query time

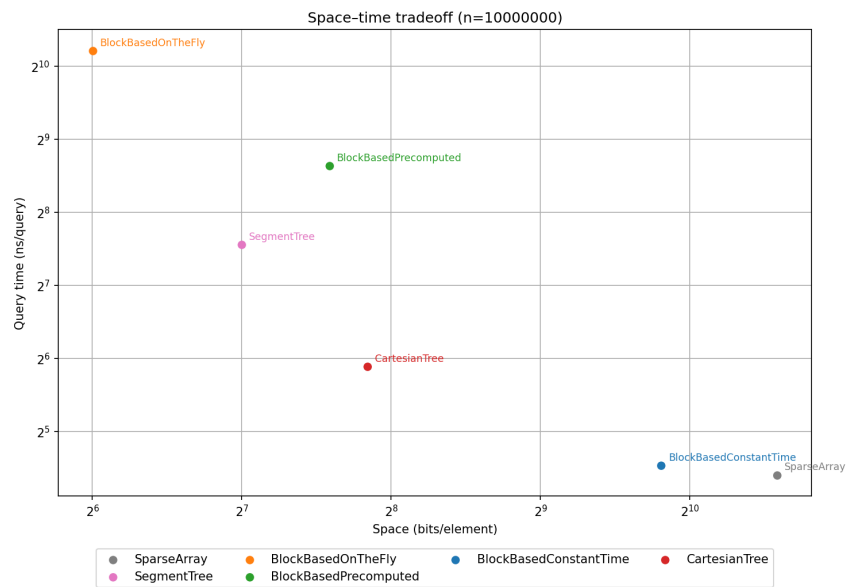


Figure 3 space-time trade-off